Integrative Structure Validation Report

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The following software was used in the production of this report:

IHMValidation Version 3.0
Python-IHM Version 2.7
MolProbity Version 4.5.2
ATSAS Version 3.2.1 (r14885)

PDB ID	9A04 | pdb_00009a04
PDB-Dev ID	PDBDEV_00000040
Structure Title	Integrative model of 5' Nucleotide excision repair complex of XPA-DBD and RPA70AB
Structure Authors	Cordoba JJ; Topolska-Wos AM; Chazin WJ
Deposited on	2019-12-19

This is a PDB-IHM Structure Validation Report.

We welcome your comments at helpdesk@pdb-ihm.org

A user guide is available at https://pdb-ihm.org/validation_help.html with specific help available everywhere you see the ? symbol.

List of references used to build this report is available here.

1. Overview

1.1. Summary

This entry consists of 1 model(s). A total of 8 dataset(s) were used to build this entry.

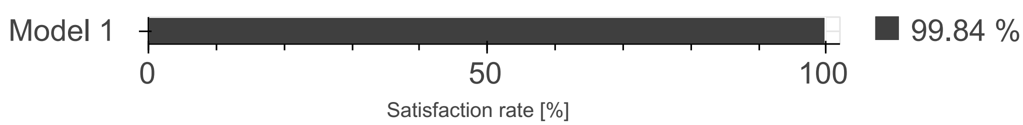
Name	Type	Count
NMR data	Experimental data	1
Other	Experimental data	3

Name	Type	Count
SAS data	Experimental data	1
Experimental model	Starting model	2
Comparative model	Starting model	1

1.2. Overall quality

This validation report contains model quality assessments for all structures, data quality and fit to model assessments for SAS and crosslinking-MS datasets. Data quality and fit to model assessments for other datasets and model uncertainty are under development. Number of plots is limited to 256.

Model Quality: Excluded Volume Analysis



2. Model Details

2.1. Ensemble information

This entry consists of 0 distinct ensemble(s).

2.2. Representation

This entry has 1 representation(s).

ID	Model(s)	Entity ID	Molecule name	Chain(s) [auth]	Total residues	Rigid segments	Flexible segments	Model coverage/ Starting model coverage (%)	Scale
1	1	1	Subunit A	A	238	1-238	-	100.00 / 100.00	Atomic
		2	Subunit B	B	142	1-142	-	100.00 / 100.00	Atomic
		3	DNA short arm	C	14	1-14	-	100.00 / 0.00	Atomic
		4	DNA long arm	D	22	1-22	-	100.00 / 0.00	Atomic

2.3. Datasets used for modeling

There are 8 unique datasets used to build the models in this entry.

ID	Dataset type	Database name	Data access code
1	NMR data	BMRB	27131
2	SAS data	SASBDB	SASDH44
3	Experimental model	PDB	pdb_00001jmc

ID	Dataset type	Database name	Data access code
4	Experimental model	PDB	pdb_00005a39
5	Comparative model	Not available	Not available
6	Other	Not available	Not available
7	Other	Not available	Not available
Not available	Other	Not available	Not available

2.4. Methodology and software

This entry is a result of 1 distinct protocol(s).

Step number	Protocol ID	Method name	Method type	Method description	Number of computed models	Multi state modeling	Multi scale modeling
1	1	Homolgy Modeling of XPA	MODELLER	Not available	10	False	False
2	1	Extension of C-terminal Helix of XPA	ROSETTA Remodel	Not available	1000	False	False
3	1	Docking of DNA to XPA	HADDOCK	Not available	1000	False	False
4	1	Randomization and Rigid Body Energy Minimization	HADDOCK	Not available	1000	False	False
5	1	Semi-flexible simulated annealing	HADDOCK	Not available	1000	False	False
6	1	Flexible explicit solvent refinement	HADDOCK	Not available	200	False	False
7	1	Stepwise addition of nucleotide linker	ROSETTA stepwise	Not available	100	False	False

There are 4 software packages reported in this entry.

ID	Software name	Software version	Software classification	Software location
1	HADDOCK	Not available	model building	http://haddock.science.uu.nl/services/HADDOCK/
2	ROSETTA	Not available	model building	https://www.rosettacommons.org/
3	MODELLER	Not available	model building	https://salilab.org/modeller/
4	FOXS	Not available	Not available	https://modbase.compbio.ucsf.edu/foxs/index.html

3. Data quality

3.1.1. Scattering profile

SAS data used in this integrative model could not be validated as the sasCIF file is currently unavailable or incomplete.

3.4. NMR

Validation for this section is under development.

4. Model quality

For models with atomic structures, MolProbity analysis is performed. For models with coarse-grained or multi-scale structures, excluded volume analysis is performed.

4.1a. Excluded Volume Analysis

Excluded volume satisfaction for the models in the entry are listed below. The Analysed column shows the number of particle-partice or particle-atom pairs for which excluded volume was analysed.

Model ID	Analysed	Number of violations	Excluded Volume Satisfaction (%)
1	11836545	18878	99.84

5. Fit to Data Used for Modeling Assessment

SAS data used in this integrative model could not be validated as the sasCIF file is currently unavailable or incomplete.

5.4. NMR

Validation for this section is under development.

6. Fit to Data Used for Validation Assessment

Validation for this section is under development.

Acknowledgments

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